Supplementary Material: Certified Residual Bounds for a Finite Gravity–Capillary Wave Profile

Abstract

We describe the computer-assisted proof of a uniform residual bound for a fixed finite Fourier approximation to a traveling gravity–capillary wave.

Contents

1 Introduction and proof strategy **2**

2 The two estimation pathways **3**

3 First term: the Path-A product $yH[y_u]$ **6**

4 Second term: the sampled estimate of $H[yy_u]$ **7**

5 Third term: local capillarity and positivity of D **8**

6 Fourth term: the nonlocal capillary estimate **10**

7 Assembling the residual and completing the proof **12**

8 Numerical choices and the reported certificate **13**

A Implementation and reproducibility **13**

B Exact data and the Lean certificate boundary **15**

1 Introduction and proof strategy

1.1 The candidate and the statement being certified

The purpose of the computer-assisted calculation is to certify the accuracy of an explicit numerical traveling-wave profile. After the search and polishing stages, the profile is frozen as the finite cosine polynomial

$$y(u) = y_N(u) = \frac{a_0}{2} + \sum_{k=1}^{N} a_k \cos(ku), \tag{1.1}$$

for some fixed integer $N \geq 1$, where the coefficients in Eq. (1.1) are fixed binary floating-point numbers. The verification treats their exact dyadic values as the definition of the candidate. The neural network is therefore a search tool, not part of the object appearing in the theorem.

We use the 2π-periodic Hilbert-transform convention

$$H[\cos(ku)] = \sin(ku), \qquad H[\sin(ku)] = -\cos(ku), \qquad H[1] = 0, \qquad k \geq 1. \tag{1.2}$$

With the sign convention of the manuscript, the residual is

$$\begin{aligned} R(y) = -\mathcal{B}(y) = \mu H[y_u] - y - yH[y_u] - H[yy_u] \\ + \beta\, \partial_u \left(\frac{y_u}{\sqrt{y_u^2 + (1 + H[y_u])^2}} \right) - \beta H \left[\partial_u \left(\frac{1 + H[y_u]}{\sqrt{y_u^2 + (1 + H[y_u])^2}} \right) \right]. \end{aligned} \tag{1.3}$$

We introduce the shorthands

$$D = \sqrt{y_u^2 + (1 + H[y_u])^2}, \qquad G = \frac{1 + H[y_u]}{D}. \tag{1.4}$$

We establish a positive lower bound for D before constructing the quotients.

For the fixed stored parameters μ and β, the purpose of the calculation is to prove

$$\boxed{D(u) \geq d_* > 0, \qquad |R(y_N)(u)| \leq \varepsilon_* \quad \text{for every } u \in \mathbb{R}.} \tag{1.5}$$

The numerical choices and resulting bounds for the proof run are collected in Section 8; their exact rational representations appear in Appendix B. This is an accuracy statement for the supplied profile. A theorem asserting an exact nearby wave would additionally require an inverse estimate for the linearized operator and a suitable existence argument.

1.2 How the finite calculation yields a continuum bound

The proof proceeds as follows.

1. Use the finite Fourier representation to evaluate y, y_u, $H[y_u]$, and their derivatives, with outward rounding.

2. Build local derivative intervals and global derivative bounds for the four nonlinear terms. Finite Hilbert transforms use Path A; sampled transforms with analytic aliasing corrections use Path B. Before forming the capillary quotients, prove $D \geq d > 0$ on the entire domain.

3. Combine the signed derivative intervals according to (1.3). Then take absolute values and apply a Taylor remainder estimate on every grid cell.

The signed assembly in the last stage preserves the cancellation between residual terms needed to obtain the small residual bound.

Julia/Arb carries out the outward-rounded numerical calculation. Lean formalizes the mathematical definitions and reusable analytic estimates. For the archived certificate, the final numerical denominator and residual bounds enter Lean through two explicit assumptions recording the external Julia/Arb calculation. The precise boundary is stated in Appendix B.4.

1.3 Common notation for interval Taylor data

Use the uniform grid and full cell width

$$u_j = jh, \qquad h = \frac{2\pi}{N_{\mathrm{grid}}}, \qquad 0 \le j < N_{\mathrm{grid}}, \qquad N_{\mathrm{grid}} \ge 4 \text{ even}. \tag{1.6}$$

The cells $[u_j, u_j + h]$ cover one period. We denote the final Taylor order by an integer $M \ge 0$. An order-q record for a function F consists of closed intervals $I_{j,m}(F)$ enclosing the derivative values at the grid nodes, together with nonnegative scalar bounds $C_m(F)$ satisfying

$$\begin{aligned} &F^{(m)}(u_j) \in I_{j,m}(F), \quad 0 \le m \le q, \\ &\left\| F^{(m)} \right\|_\infty \le C_m(F), \quad 0 \le m \le q+1. \end{aligned} \tag{1.7}$$

All norms are over a complete period, equivalently over $\mathbb{R}$ for periodic functions. The intervals $I_{j,m}(F)$ enclose raw derivatives; the factor $1/m!$ is introduced in the Taylor expansion. We also write

$$U_{j,m}(F) = \sup\{|x| : x \in I_{j,m}(F)\}. \tag{1.8}$$

For an Arb ball $[c \pm r] = [c - r, c + r]$, this is $|c| + r$.

Arb evaluates elementary arithmetic operations, trigonometric functions, and admissible square roots and reciprocals with outward rounding. The working precision is specified with the run parameters in Section 8. A symbol $C_m(F)$ always denotes a nonnegative scalar upper bound; if its defining expression is computed as a ball, the outward upper endpoint is used.

Pointwise interval arithmetic accounts for rounding error. Additional analytic estimates account for variation between nodes, unresolved Fourier modes, and derivative growth. These estimates are described explicitly below and are also evaluated with outward rounding.

2 The two estimation pathways

2.1 Path A: finite Fourier algebra

For a finite trigonometric polynomial

$$F(u) = \frac{\alpha_0}{2} + \sum_{k=1}^{J} \big(\alpha_k \cos(ku) + \gamma_k \sin(ku)\big), \tag{2.1}$$

the Hilbert transform is the finite expression

$$H[F](u) = \sum_{k=1}^{J} \big(\alpha_k \sin(ku) - \gamma_k \cos(ku)\big). \tag{2.2}$$

Differentiation and multiplication also have finite coefficient formulas. The finite representation retains every generated Fourier mode. Arb encloses rounding in coefficient arithmetic and evaluation, and products retain the degree they generate.

Write $H_y = H[y_u]$. Applied to the candidate, Path A gives

$$y_u(u) = -\sum_{k=1}^{N} k a_k \sin(ku), \qquad H_y(u) = \sum_{k=1}^{N} k a_k \cos(ku). \tag{2.3}$$

The latter expression is used in the product yH_y and in the capillary denominator D.

Finite coefficient sums provide global derivative bounds:

$$\left\| F^{(m)} \right\|_\infty \le \sum_{k=1}^{J} k^m (|\alpha_k| + |\gamma_k|), \qquad m \ge 1, \tag{2.4}$$

with $|\alpha_0|/2$ included for $m = 0$. In particular, usable seed bounds for the candidate are

$$C_0(y) = \frac{|a_0|}{2} + \sum_{k=1}^{N} |a_k|, \qquad C_m(y) = \sum_{k=1}^{N} k^m |a_k| \quad (m \ge 1), \tag{2.5}$$

$$C_m(y_u) = \sum_{k=1}^{N} k^{m+1} |a_k|, \qquad C_m(H_y) = \sum_{k=1}^{N} k^{m+1} |a_k| \quad (m \ge 0). \tag{2.6}$$

Evaluating the differentiated finite series at the nodes supplies $I_{j,m}$ directly.

2.2 Path B: interval data and a certified sampled transform

The capillary quotients y_u/D and G generally have infinitely many Fourier modes even though y is a finite polynomial. Path B works from interval-enclosed samples and derivatives, and corrects a discrete Hilbert transform to enclose the continuous one. Its nonlinear input data are constructed by interval Taylor arithmetic.

Let F be a smooth periodic function. For even N_{grid}, the discrete Hilbert transform has multipliers $-i$ on positive principal frequencies, $+i$ on negative principal frequencies, and zero on the constant and Nyquist ($N_{\text{grid}}/2$) modes. At a node it can be written as

$$H_{N_{\text{grid}}}[F](u_j) = \sum_{n=0}^{N_{\text{grid}}-1} h_n F(u_{j-n \bmod N_{\text{grid}}}), \qquad h_n = \begin{cases} \dfrac{2}{N_{\text{grid}}} \cot\left(\dfrac{\pi n}{N_{\text{grid}}}\right), & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases} \tag{2.7}$$

This kernel is the inverse discrete Fourier transform of the multiplier. The circular convolution is computed by multiplying polynomials whose coefficients are the kernel and sampled values, then wrapping the resulting coefficient indices modulo N_{grid}. The convolution is evaluated using interval arithmetic with outward rounding, so the resulting intervals contain the exact discrete Hilbert-transform values.

Correction for aliasing. Write

$$\widehat{F}_k = \frac{1}{2\pi} \int_0^{2\pi} F(u) e^{-iku}\, du. \tag{2.8}$$

For an integer $p \geq 2$, periodic integration by parts gives

$$|\widehat{F}_k| \leq \frac{C_p(F)}{|k|^p}, \qquad k \neq 0. \tag{2.9}$$

Sampling identifies modes whose indices differ by a multiple of N_{grid}. For $|k| < N_{\mathrm{grid}}/2$ the continuous and discrete multipliers agree; for the remaining modes their difference has absolute value at most two. Consequently, for even $N_{\mathrm{grid}} \geq 4$,

$$\begin{aligned} |H[F](u_j) - H_{N_{\mathrm{grid}}}[F](u_j)| &\leq 2 \sum_{|k| \geq N_{\mathrm{grid}}/2} |\widehat{F}_k| \\ &\leq 4C_p(F) \sum_{k=N_{\mathrm{grid}}/2}^{\infty} k^{-p} \\ &\leq \frac{4C_p(F)}{(p-1)(N_{\mathrm{grid}}/2-1)^{p-1}} =: \varepsilon_{\mathrm{alias}}(F;p). \end{aligned} \tag{2.10}$$

The last inequality follows by bounding the decreasing sum by the integral $\int_{N_{\mathrm{grid}}/2-1}^{\infty} x^{-p}\,dx$. Adding the symmetric interval $[-\varepsilon_{\mathrm{alias}}(F;p), \varepsilon_{\mathrm{alias}}(F;p)]$ to the discrete output therefore encloses the continuous transform at each node.

Derivative records for the transformed function. For the smooth functions used here, $\partial_u^m H[F] = H[F^{(m)}]$. Let $\mathcal{H}_{j,m}(F)$ be the interval obtained by applying (2.7) to the entire sample column $\{I_{\ell,m}(F)\}_{\ell=0}^{N_{\mathrm{grid}}-1}$. Then

$$I_{j,m}(H[F]) = \mathcal{H}_{j,m}(F) + [-e_m(F), e_m(F)], \qquad e_m(F) = \frac{4C_{m+p}(F)}{(p-1)(N_{\mathrm{grid}}/2-1)^{p-1}} \tag{2.11}$$

is valid whenever the indicated global derivative bound is available. One may choose a separate admissible $p \geq 2$ for each column; the notation above displays the rule for any such choice.

A global bound follows from the same Fourier decay with $p = 2$:

$$\|H[F]\|_\infty \leq 2\sum_{k=1}^{\infty} k^{-2} \left\|F''\right\|_\infty = \frac{\pi^2}{3} \left\|F''\right\|_\infty. \tag{2.12}$$

Thus an admissible output bound at derivative order m is

$$C_m(H[F]) = \frac{\pi^2}{3} C_{m+2}(F). \tag{2.13}$$

The extra derivatives are part of the proof: a final order-M transformed record needs more input information than its $M+1$ sampled columns alone. The two uses of Path B below spell out these requirements.

2.3 Products, nonlinear operations, and tightening

Both pathways use the exact Leibniz identity to form product records:

$$I_{j,m}(FV) = \sum_{r=0}^{m} \binom{m}{r} I_{j,r}(F) I_{j,m-r}(V), \tag{2.14}$$

$$C_m^{\mathrm{seed}}(FV) = \sum_{r=0}^{m} \binom{m}{r} C_r(F) C_{m-r}(V). \tag{2.15}$$

The first line uses signed interval arithmetic; the second is a nonnegative upper bound. Square roots and reciprocals use composition formulas on intervals with strictly positive lower endpoints. The denominator and quotient identities in Section 5 give explicit bounds for their higher derivatives.

A record also permits downward tightening. Define $V_m(F) = \max_j U_{j,m}(F)$. The fundamental theorem of calculus on a cell implies

$$\left\|F^{(m)}\right\|_\infty \le V_m(F) + hC_{m+1}(F). \tag{2.16}$$

For an order-q record, retain C_{q+1} and update successively

$$C_m(F) \leftarrow \min\{C_m(F),\, V_m(F) + hC_{m+1}(F)\}, \qquad m = q, q-1, \ldots, 0. \tag{2.17}$$

Each candidate is rounded upward, and the already tightened next-order bound is used. The pass leaves the local intervals and top global bound unchanged. It can be useful when the improved bounds feed a later product, composition, or aliasing correction.

The four estimates now follow in the order

$$yH[y_u] \quad \longrightarrow \quad H[yy_u] \quad \longrightarrow \quad \partial_u(y_u/D) \quad \longrightarrow \quad H[\partial_u G]. \tag{2.18}$$

The first term uses Path A. The remaining three terms use Path B.

3 First term: the Path-A product $yH[y_u]$

3.1 Constructing the local product record

Set $P = yH_y$. Both factors are explicit finite cosine polynomials, and P has degree at most $2N$. Path A evaluates H_y by the finite coefficient formula before multiplication by y. The product retains its full degree.

For $m \ge 1$, the local derivatives of y are enclosed by evaluating

$$y^{(m)}(u_j) = \sum_{k=1}^{N} k^m a_k \cos\!\left(ku_j + \frac{m\pi}{2}\right). \tag{3.1}$$

For $m \ge 0$, the corresponding formula for the other factor is

$$H_y^{(m)}(u_j) = \sum_{k=1}^{N} k^{m+1} a_k \cos\!\left(ku_j + \frac{m\pi}{2}\right). \tag{3.2}$$

The zeroth derivative of y includes $a_0/2$ as in (1.1). Using these two records, form

$$\boxed{I_{j,m}(P) = \sum_{r=0}^{m} \binom{m}{r} I_{j,r}(y) I_{j,m-r}(H_y), \qquad 0 \le m \le M.} \tag{3.3}$$

For example,

$$P' = y'H_y + yH'_y, \tag{3.4}$$

$$P'' = y''H_y + 2y'H'_y + yH''_y. \tag{3.5}$$

The exact Leibniz identity proves that each interval contains the required derivative. Interval arithmetic preserves this inclusion as dependencies between inputs contribute to the enclosure width.

3.2 The remainder derivative and the whole-cell enclosure

The coefficient bounds in (2.5)–(2.6) give the global remainder constant

$$\boxed{C_{M+1}(P) = \sum_{r=0}^{M+1} \binom{M+1}{r} C_r(y) C_{M+1-r}(H_y).} \tag{3.6}$$

It controls $P^{(M+1)}$ at every point, including between nodes. The final order-M product record combines local derivatives through order M with this global bound on the derivative of order $M+1$.

For $u = u_j + t$ with $0 \le t \le h$, Taylor's formula gives

$$P(u_j + t) = \sum_{m=0}^{M} \frac{P^{(m)}(u_j) t^m}{m!} + \frac{1}{M!} \int_0^t (t-s)^M P^{(M+1)}(u_j + s)\, ds. \tag{3.7}$$

The remainder has absolute value at most $C_{M+1}(P) t^{M+1}/(M+1)!$. Hence

$$\boxed{P(u_j + t) \in \sum_{m=0}^{M} I_{j,m}(P) \frac{t^m}{m!} + \left[-\frac{C_{M+1}(P) t^{M+1}}{(M+1)!}, \frac{C_{M+1}(P) t^{M+1}}{(M+1)!} \right].} \tag{3.8}$$

In particular,

$$\|P\|_\infty \le \max_j \left\{ \sum_{m=0}^{M} U_{j,m}(P) \frac{h^m}{m!} + C_{M+1}(P) \frac{h^{M+1}}{(M+1)!} \right\}. \tag{3.9}$$

This illustrates the grid-to-continuum argument. Residual assembly uses the signed intervals $I_{j,m}(P)$ with the minus sign in (1.3).

The product's downward tightening pass can improve $C_0(P), \ldots, C_M(P)$. Its signed local intervals and global remainder bound $C_{M+1}(P)$ remain unchanged, so its additive contribution to the final residual scan is preserved. Within a chain of nonlinear operations, tightened lower-order bounds can improve subsequent estimates.

4 Second term: the sampled estimate of $H[yy_u]$

4.1 Forming the input before applying the transform

Let $F = yy_u$. Construct its signed derivative intervals by

$$I_{j,m}(F) = \sum_{r=0}^{m} \binom{m}{r} I_{j,r}(y) I_{j,m-r}(y_u). \tag{4.1}$$

The corresponding global bounds are

$$C_m(F) = \sum_{r=0}^{m} \binom{m}{r} C_r(y) C_{m-r}(y_u). \tag{4.2}$$

Equivalently, one may substitute $(y_u)^{(s)} = y^{(s+1)}$. This makes explicit that forming the product at a given derivative order requires one more derivative of y in the y_u factor. All these input bounds come from finite coefficient sums and, where used, valid tightening.

4.2 Transforming each derivative column

The implemented calculation sends F through Path B. For $T = H[F]$ and $0 \le m \le M$, equations (2.7) and (2.11) give

$$\boxed{I_{j,m}(T) = \mathcal{H}_{j,m}(F) + [-e_m(F), e_m(F)], \qquad e_m(F) = \frac{4C_{m+p}(F)}{(p-1)(N_{\mathrm{grid}}/2-1)^{p-1}}.} \tag{4.3}$$

The convolution uses the full grid column, so this is a global operation even though its result is a derivative interval at one node. Its two error sources are accounted for separately: the convolution balls enclose numerical rounding and input uncertainty, while $e_m(F)$ bounds the continuous-to-discrete difference.

For the final Taylor remainder, (2.13) yields

$$\boxed{C_{M+1}(T) = \frac{\pi^2}{3} C_{M+3}(F).} \tag{4.4}$$

The local Taylor polynomial uses derivatives through order M, and the global remainder estimate requires a bound on the derivative of F of order $M+3$. The aliasing correction for column m uses $C_{m+p}(F)$ for the chosen decay order p.

Why use Path B for a finite polynomial? The product yy_u is a trigonometric polynomial of degree at most $2N$ and admits a direct Path-A transform. When $N_{\mathrm{grid}} > 4N$, its exact sampled transform agrees with the continuous transform at every node. The archived implementation uses the generic Path-B routine and its conservative correction for this term, providing a common computational procedure for both $H[yy_u]$ and the final capillary transform. For the capillary transform, the correction controls the infinite Fourier tail of its nonlinear input.

5 Third term: local capillarity and positivity of D

The local capillary term is

$$L = \partial_u\left(\frac{y_u}{D}\right) = Q', \qquad Q = \frac{y_u}{D}. \tag{5.1}$$

Its difficulty is the nonlinear denominator. The term is constructed using nonlinear interval Taylor arithmetic and the Path-A expression for $H[y_u]$. The construction starts by proving that division by D is valid everywhere.

5.1 A Lipschitz lower bound for the denominator

For the denominator estimates, set

$$A = y_u, \qquad B = 1 + H[y_u], \qquad D = \sqrt{A^2 + B^2}. \tag{5.2}$$

The Path-A bounds give

$$\begin{aligned} C_m(A) &= C_m(y_u), && && m \ge 0, \\ C_0(B) &= 1 + C_0(H_y), && C_m(B) = C_m(H_y), && m \ge 1. \end{aligned} \tag{5.3}$$

At each node, evaluate A, B, and $\sqrt{A^2+B^2}$ using outward rounding, and let $\underline{D}_j$ be a lower endpoint for $D(u_j)$. At this stage the square root evaluates the Euclidean norm of (A, B). The interval squares are restricted to their known nonnegative range.

Proposition 5.1. *The first-derivative bounds give*

$$D(u) \geq d_{\mathrm{Lip}} := \min_{0 \leq j < N_{\mathrm{grid}}} \underline{D}_j - h(C_1(A) + C_1(B)) \quad \textit{for all } u \in \mathbb{R}. \tag{5.4}$$

In particular, a strictly positive outward-rounded lower bound for d_{Lip} proves that D is bounded away from zero.

Proof. The reverse triangle inequality in $\mathbb{R}^2$ gives

$$\begin{aligned} |D(u) - D(v)| &\leq \sqrt{(A(u) - A(v))^2 + (B(u) - B(v))^2} \\ &\leq |A(u) - A(v)| + |B(u) - B(v)| \\ &\leq (C_1(A) + C_1(B))|u - v|. \end{aligned} \tag{5.5}$$

For $u = u_j + t$, $0 \leq t \leq h$, we obtain $D(u) \geq \underline{D}_j - h(C_1(A) + C_1(B))$. Taking the minimum over the cells and using periodicity proves (5.4). □

The bounds $C_1(A)$ and $C_1(B)$ are available from Path A. The argument establishes positivity using bounds on A, B, and their first derivatives. Reciprocal evaluation proceeds once the computed lower bound is strictly positive.

After positivity is established, D is smooth. First construct its derivative bounds using $d_{\mathrm{Lip}} > 0$ in the recurrences below. These already validated records then give the additional Taylor lower bound

$$d_{\mathrm{Tay}} = \min_j \left\{ \underline{D}_j - \sum_{m=1}^{M} U_{j,m}(D) \frac{h^m}{m!} - C_{M+1}(D) \frac{h^{M+1}}{(M+1)!} \right\}. \tag{5.6}$$

All subtractions in this expression are rounded in the lower-bound direction. Taking $d = \max\{d_{\mathrm{Lip}}, d_{\mathrm{Tay}}\}$ retains a valid positive lower bound. Subsequent derivative and quotient bounds may then be recomputed using this improved d. The Taylor estimate thus refines the initial Lipschitz lower bound.

5.2 Derivative bounds for the denominator

Set $S = A^2 + B^2 = D^2$. The product rule supplies local intervals for S and seed bounds

$$C_m(S) = \sum_{r=0}^{m} \binom{m}{r} (C_r(A) C_{m-r}(A) + C_r(B) C_{m-r}(B)). \tag{5.7}$$

Since S is a finite trigonometric polynomial, its coefficient representation gives another available bound. For $m \geq 1$, differentiating $D^2 = S$ gives

$$2DD^{(m)} = S^{(m)} - \sum_{r=1}^{m-1} \binom{m}{r} D^{(r)} D^{(m-r)}. \tag{5.8}$$

The sum is empty when $m = 1$. Starting with $C_0(D) = \sqrt{C_0(A)^2 + C_0(B)^2}$, one may use

$$\boxed{C_m(D) = \frac{1}{2d} \left(C_m(S) + \sum_{r=1}^{m-1} \binom{m}{r} C_r(D) C_{m-r}(D) \right).} \tag{5.9}$$

Indeed, divide (5.8) by $2D$, use $D \geq d > 0$, and take supremum bounds. This proves the recurrence inductively.

At a node, the same identity can be evaluated with signed intervals:

$$I_{j,m}(D) = \frac{I_{j,m}(S) - \sum_{r=1}^{m-1} \binom{m}{r} I_{j,r}(D) I_{j,m-r}(D)}{2 I_{j,0}(D)}. \tag{5.10}$$

The zeroth interval is restricted using the certified lower bound d, so its denominator excludes zero. Equation (5.10) gives an explicit recursive enclosure; the implementation also constructs local data through square-root composition formulas. Both constructions follow from the same exact identities and interval inclusion principle.

5.3 The quotient and the local capillary derivative

The identity $QD = A$ yields, for $m \geq 1$,

$$Q^{(m)} D = A^{(m)} - \sum_{r=0}^{m-1} \binom{m}{r} Q^{(r)} D^{(m-r)}. \tag{5.11}$$

For example,

$$Q' = \frac{A'}{D} - \frac{AD'}{D^2}. \tag{5.12}$$

Begin with $I_{j,0}(Q) = I_{j,0}(A)/I_{j,0}(D)$ and use

$$I_{j,m}(Q) = \frac{I_{j,m}(A) - \sum_{r=0}^{m-1} \binom{m}{r} I_{j,r}(Q) I_{j,m-r}(D)}{I_{j,0}(D)} \tag{5.13}$$

as an explicit signed enclosure. Corresponding global bounds are

$$\begin{aligned} C_0(Q) &= \frac{C_0(A)}{d}, \\ C_m(Q) &= \frac{1}{d} \left(C_m(A) + \sum_{r=0}^{m-1} \binom{m}{r} C_r(Q) C_{m-r}(D) \right), \qquad m \geq 1. \end{aligned} \tag{5.14}$$

Alternatively, the quotient record can be constructed as the product of A with a reciprocal record for D. In the implementation, reciprocals and square roots are propagated by composition formulas; the algebraic identities expose why the resulting estimates require a positive d. Different rigorous constructions may have different widths.

Finally, differentiation shifts the record by one order:

$$\boxed{I_{j,m}(L) = I_{j,m+1}(Q) \quad (0 \leq m \leq M), \qquad C_{M+1}(L) = C_{M+2}(Q).} \tag{5.15}$$

Thus the quotient must be constructed through local order $M+1$ and supplied with a global derivative bound of order $M+2$. The contribution to the residual is $+\beta I_{j,m}(L)$, with coefficient $|\beta|$ in its global bound on the derivative of order $M+1$.

6 Fourth term: the nonlocal capillary estimate

The final term is $H[G']$ with $G = B/D$. It combines the nonlinear quotient construction with a sampled Hilbert transform. It therefore requires every ingredient developed for the preceding terms.

6.1 Derivative data for the nonlinear input

Once $D \geq d > 0$ is certified, G is smooth and periodic. The identity $GD = B$ gives

$$G^{(m)}D = B^{(m)} - \sum_{r=0}^{m-1} \binom{m}{r} G^{(r)} D^{(m-r)}, \qquad m \geq 1. \tag{6.1}$$

Start with $I_{j,0}(G) = I_{j,0}(B)/I_{j,0}(D)$. The signed interval version of (6.1) has exactly the form (5.13), with A, Q replaced by B, G. The corresponding global bounds are

$$\begin{aligned} C_0(G) &= \frac{C_0(B)}{d}, \\ C_m(G) &= \frac{1}{d}\left(C_m(B) + \sum_{r=0}^{m-1} \binom{m}{r} C_r(G) C_{m-r}(D) \right), \qquad m \geq 1. \end{aligned} \tag{6.2}$$

The source calculation compares algebraic D/G derivative bounds with composition-based bounds and retains the tighter valid upper endpoints where needed. These estimates apply directly to the smooth functions D and G.

Differentiation supplies the input to Path B:

$$I_{j,m}(G') = I_{j,m+1}(G), \qquad C_m(G') = C_{m+1}(G). \tag{6.3}$$

6.2 The transform, aliasing correction, and remainder bound

Apply the certified convolution to each derivative column of G'. For $0 \leq m \leq M$, the output record is

$$\boxed{\begin{aligned} I_{j,m}(H[G']) &= \mathcal{H}_{j,m}(G') + [-e_m(G'), e_m(G')], \\ e_m(G') &= \frac{4C_{m+p+1}(G)}{(p-1)(N_{\mathrm{grid}}/2-1)^{p-1}}. \end{aligned}} \tag{6.4}$$

Here $\mathcal{H}_{j,m}(G')$ uses the samples of $G^{(m+1)}$. Since G generally has infinitely many Fourier modes, the correction bounds the contribution of its unresolved modes to the transform error.

For the final remainder we use

$$\left(H[G']\right)^{(M+1)} = H[G^{(M+2)}], \qquad \boxed{C_{M+1}(H[G']) = \frac{\pi^2}{3} C_{M+4}(G).} \tag{6.5}$$

This shows why high derivatives of the quotient matter. An order-M residual record requires local derivatives of G through order $M+1$, a global derivative bound of order $M+4$ for (6.5), and any further bounds required by the selected aliasing decay orders. For example, $p = 2$ would use $C_{m+3}(G)$ for the local corrections, through order $M+3$. This example illustrates how the derivative requirements depend on the chosen decay order p.

The contribution to the residual is $-\beta I_{j,m}(H[G'])$. Its global contribution to the derivative bound of order $M+1$ is $|\beta| C_{M+1}(H[G'])$.

6.3 Derivative information required by the four terms

Table 1 collects the input needed for a final order-M record. Bounds used in aliasing corrections are additional to those in the last column and follow (4.3) and (6.4).

Term	Local input derivatives	Bound for derivative $M+1$
$P = yH_y$	y, H_y through order M	$\sum_{r=0}^{M+1} \binom{M+1}{r} C_r(y) C_{M+1-r}(H_y)$
$T = H[yy_u]$	yy_u through order M	$\frac{\pi^2}{3} C_{M+3}(yy_u)$
$L = Q'$	Q through order $M+1$	$C_{M+2}(Q)$
$H[G']$	G through order $M+1$	$\frac{\pi^2}{3} C_{M+4}(G)$

Table 1: Derivative bookkeeping for the four nonlinear residual terms.

7 Assembling the residual and completing the proof

7.1 Signed assembly at each node

The linear terms $\mu H_y - y$ already have Path-A records. For each node and $0 \le m \le M$, form the interval

$$\begin{aligned} I_{j,m}(R) = \mu I_{j,m}(H_y) - I_{j,m}(y) - I_{j,m}(P) - I_{j,m}(T) \\ + \beta I_{j,m}(L) - \beta I_{j,m}(H[G']). \end{aligned} \tag{7.1}$$

All parameters in this expression are the exact stored values, enclosed as needed by interval arithmetic. Linearity and the previously proved inclusions give $R^{(m)}(u_j) \in I_{j,m}(R)$.

After forming (7.1), we take $U_{j,m}(R) = \sup\{|x| : x \in I_{j,m}(R)\}$. Individual terms may be much larger than the final residual; their signed combination is essential. Separately, a global remainder constant is obtained by the triangle inequality:

$$\begin{aligned} C_{M+1}(R) = |\mu| C_{M+1}(H_y) + C_{M+1}(y) + C_{M+1}(P) + C_{M+1}(T) \\ + |\beta| (C_{M+1}(L) + C_{M+1}(H[G'])). \end{aligned} \tag{7.2}$$

The triangle inequality supplies a global remainder bound, which is multiplied by $h^{M+1}/(M+1)!$ in the final estimate.

7.2 From the finite grid to every real argument

For $u = u_j + t$ with $0 \le t \le h$, Taylor's theorem and (7.2) imply

$$|R(u)| \le \sum_{m=0}^{M} U_{j,m}(R) \frac{h^m}{m!} + C_{M+1}(R) \frac{h^{M+1}}{(M+1)!}. \tag{7.3}$$

Every point of one period belongs to a cell, so the outward-rounded maximum provides the certificate

$$\boxed{\|R(y_N)\|_\infty \le \varepsilon_* := \max_{0 \le j < N_{\text{grid}}} \left\{ \sum_{m=0}^{M} U_{j,m}(R) \frac{h^m}{m!} + C_{M+1}(R) \frac{h^{M+1}}{(M+1)!} \right\}.} \tag{7.4}$$

Periodicity extends the conclusion to all $u \in \mathbb{R}$. Together with the positive lower bound from Section 5.1, this completes the mathematical implication from validated interval records and global bounds to the two continuum inequalities.

The cell width in (7.4) is $h = 2\pi/N_{\text{grid}}$. For fixed $C_{M+1}(R)$, the remainder factor scales as $N_{\text{grid}}^{-(M+1)}$. The full numerical bound also includes local interval widths and aliasing allowances.

8 Numerical choices and the reported certificate

The archived calculation uses

$$N = 150, \qquad N_{\mathrm{grid}} = 2{,}000{,}000, \qquad M = 6, \qquad p_{\mathrm{Arb}} = 128 \text{ bits}, \tag{8.1}$$

with grid shards of 1024 points. The parameter displays are

$$\mu \approx 1.3660618372350841, \qquad \beta \approx 0.248004. \tag{8.2}$$

For this choice $M = 6$, the final Taylor remainder has order 7. The transform of yy_u uses a global derivative bound of order 9 for its remainder estimate; the local capillary quotient needs order 8, and the nonlocal capillary quotient G needs order 10. These are the numerical instances of $M + 3$, $M + 2$, and $M + 4$, respectively, in the preceding sections. The certified result is

$$\boxed{D(u) \geq 0.3446588, \qquad |R(y_{150})(u)| \leq 1.78 \times 10^{-13} \quad \text{for every } u \in \mathbb{R}.} \tag{8.3}$$

Here p_{Arb} is the arithmetic precision and is distinct from the Fourier-decay order p in the aliasing estimate. The reported output of the supremum-bound computation is a ball B_R with decimal displays

$$\mathrm{mid}(B_R) \approx 1.9928311263803702 \times 10^{-14}, \tag{8.4}$$

$$\mathrm{rad}(B_R) \approx 1.5799937398052128 \times 10^{-13}, \tag{8.5}$$

$$\inf B_R \approx -1.3807106271671758 \times 10^{-13}, \tag{8.6}$$

$$\sup B_R \approx 1.7792768524432498 \times 10^{-13}. \tag{8.7}$$

The ball B_R is the output of the supremum-bound computation. Its exact outward upper endpoint supplies the bound for $\|R\|_\infty$ and is recorded in Appendix B. The midpoint and radius describe this bound-expression ball.

The denominator endpoint is approximately 0.34465883518530396. Exact rational comparisons give the outward-safe bounds in (8.3). The residual certificate quantifies the assembled expression containing P, T, L, and $H[G']$.

The phrase "near machine precision" refers here to the certified total residual bound 1.78×10^{-13}. For reference, IEEE double precision has machine epsilon $2^{-52} \approx 2.22 \times 10^{-16}$, and the verifier uses 128-bit ball arithmetic. The verification precision controls arithmetic enclosures. The final bound depends on the fixed candidate and the arithmetic, Taylor, and aliasing allowances.

A Implementation and reproducibility

A.1 Correspondence with the Julia/Arb calculation

The numerical package is `BabenkoRigorous`. The mathematical construction above corresponds to the following components, as described in the supplied implementation account.

`FourierSeries.jl`. Finite trigonometric representations, differentiation, Path-A Hilbert transforms, multiplication, evaluation, and coefficient-based global bounds.

`TaylorSeries.jl`. Local derivative intervals and global bounds, Leibniz products, reciprocal and square-root composition, tightening, denominator positivity, and algebraic bounds for derivatives of D and G.

`GDerivatives.jl`. Construction of the capillary function $G = (1 + H[y_u])/D$ from the certified denominator and reciprocal data.

`FFTHilbert.jl`. Certified cotangent-kernel convolution, the analytic aliasing allowance, derivative commutation, and the global Hilbert bound.

`ResidualBound.jl`. Assembly of the four nonlinear terms with the linear terms, followed by the Taylor scan in (7.4). Both $H[yy_u]$ and $H[G']$ pass through `fft_hilbert_series` in the implemented route.

`ShardStore.jl` and `Sharded.jl`. Storage and processing of the large grid in exact interval shards.

`Certificate.jl`. Export of the stored inputs and outward endpoints as exact rational literals in a generated Lean data module.

For example, the local product in Section 3 is built from `yu = deriv(y)`, `Hyu = hilbert(yu)`, and the multiplication of the corresponding Taylor records. The local entries `derivs[j,m+1]` store raw derivative intervals; the global entries `sup_bounds[m+1]` store bound expressions. Julia uses one-based array indices, whereas the grid in (1.6) is indexed from zero. The mathematics is unchanged by that indexing choice.

The quotient recurrences displayed in the main text give explicit mathematical enclosure formulas. The source account describes local square-root and reciprocal propagation by Faà di Bruno composition formulas and supplementary algebraic bounds for high derivatives of D and G. Interval widths depend on the chosen valid construction. The archived numerical endpoints refer to the implemented construction, including its tightening and sharding choices.

A.2 Sharding the two-million-point grid

The sharded schedule computes coefficient bounds once, streams local nonlinear operations over contiguous blocks, and processes each global Hilbert transform one derivative column at a time. The final residual is then reassembled shard by shard. Grid maxima used in tightening are merged over the complete grid to obtain bounds valid throughout a full period.

Intermediate Arb balls are saved and restored through exact representations of their mantissas, exponents, and radii. The source account reports that the reduction order matches the monolithic calculation and that smaller-grid tests compare the two routes. The archived run used eight workers with four Julia threads each for its initial stage and eight threads for the main transform and final stages. Separate processes between major stages released memory and left exact state dumps for subsequent stages.

Reproducibility. The input coefficients, archived run evidence, and certificate files are available in the paper repository (revision `b5a40c9`). The archived-run documentation provides artifact locations and archive checksums; the repository's Julia and Lean guides give reproduction and build instructions.

B Exact data and the Lean certificate boundary

B.1 Exact parameters and outward endpoints

The generated certificate stores 151 exact rational coefficients, the exact parameters, and the grid and Taylor settings. The parameters are

$$\mu_{\mathrm{cert}} = \frac{1538048895284245}{1125899906842624}, \qquad \beta_{\mathrm{cert}} = \frac{2233821443972785}{9007199254740992}. \tag{B.1}$$

The certificate endpoints are

$$\varepsilon_{0,\mathrm{cert}} = \frac{66570650447636094343693612891855023841}{37414441915671114706014331717536845303191873100185 6}, \tag{B.2}$$

$$D_{\mathrm{lower,cert}} = \frac{30024018999569627600479976468787845960585}{87112285931760246646623899502532662132736}. \tag{B.3}$$

These exact values satisfy

$$\varepsilon_{0,\mathrm{cert}} < \frac{178}{10^{15}} = 1.78 \times 10^{-13}, \qquad D_{\mathrm{lower,cert}} > \frac{3446588}{10^{7}} = 0.3446588. \tag{B.4}$$

The shortened decimals in (8.2) and (8.7) display approximations to these exact rational definitions.

B.2 Aligning the constant mode and the residual sign

Julia evaluates the first cosine coefficient as $a_0/2$. The generic Lean profile uses its zeroth stored entry directly. The exported vector is therefore

$$\widetilde{a}_0 = a_0/2, \qquad \widetilde{a}_k = a_k \quad (1 \le k \le N). \tag{B.5}$$

The formal convention adapter makes the two profile definitions agree and preserves the derivatives and auxiliary functions. The encoded coefficient data consist of the converted vector in the Lean convention. The source account reports a certificate-specific check of its zeroth-entry identity.

The sign bridge is $R = -\mathcal{B}$ as in (1.3), hence $|R| = |\mathcal{B}|$. The convention and sign identities are independent of the numerical denominator and residual assumptions below.

B.3 Reusable analytic results

The Lean project `WaveCAP` defines the periodic Hilbert transform by its Fourier multiplier $-i\,\mathrm{sgn}(k)$, with zero multiplier at $k = 0$, and defines the finite profile, D, G, and the residual. Its reusable library follows the same proof architecture:

- `HilbertOnTrigPoly.lean` establishes the finite-mode rules and the equality between the abstract transform of y_u and H_y.
- `AliasingErrorBound.lean`, `FourierTailHilbert.lean`, `HilbertCommutesDeriv.lean`, and `FFTHilbertCorrectness.lean` supply the sampled-transform, decay, tail, commutation, and correctness results used in Path B.
- `LeibnizBound.lean`, `FaaDiBrunoBound.lean`, `TaylorSeriesCorrectness.lean`, and `SeriesCorrectness.lean` cover derivative bounds and preservation of the Taylor-record invariant.

- `DminBoundCorrectness.lean` and `TightDGSupBounds.lean` cover positivity and the capillary derivative bounds.
- `TightenSupBoundsLemma.lean`, `TaylorRemainder.lean`, `GridToSupNormTaylor.lean`, and `ResidualAssemblyCorrectness.lean` justify tightening, residual assembly, the continuum estimate, and periodic extension.

These generic mathematical theorems establish the analytic results and data-structure contracts used by the verification scheme. Their connection to the reported numerical certificate is expressed by the two assumptions below.

B.4 The two numerical assumptions

The full calculation of the local derivative matrices, global bound vectors, and intermediate convolution enclosures is carried out in Julia/Arb. The Lean proof starts from the final Arb conclusions, imported through the certificate-specific interface as two explicit numerical assumptions. In mathematical form, they are

$$\mathsf{A}_D: \quad D_{\mathrm{lower,cert}} > 0 \quad \text{and} \quad \forall u \in \mathbb{R}, \quad D_{\mathrm{lower,cert}} \le D(a_{\mathrm{cert}}; u), \tag{B.6}$$

$$\mathsf{A}_R: \quad \mathsf{A}_D \Longrightarrow \forall u \in [0, 2\pi], \quad |R(a_{\mathrm{cert}}, \mu_{\mathrm{cert}}, \beta_{\mathrm{cert}}; u)| \le \varepsilon_{0,\mathrm{cert}}. \tag{B.7}$$

Here a_{cert} denotes the converted coefficient vector. The Lean names are

`cert_arb_dmin_bound`, `cert_arb_sup_bound_le_eps0`.

The generated data module contains rational literals; the two assumptions are declared separately in the certificate-specific axiom module.

From these assumptions Lean obtains the denominator bound, applies the residual estimate on one period, proves periodicity and extends the estimate to $\mathbb{R}$, checks (B.4) by exact rational normalization, and transfers the result to $|\mathcal{B}|$. Thus the result is a computer-assisted residual certificate with a Lean-checked conclusion conditional on two externally computed numerical anchors. The numerical trust boundary remains the Julia/Arb calculation and its implementation of the analytic estimates.

The source account reports that the residual theorem's axiom audit lists these two numerical anchors together with the standard Mathlib principles `propext`, `Classical.choice`, and `Quot.sound`. The denominator theorem uses the denominator anchor. The convention and sign-alignment results are proved independently of the certificate-specific numerical assumptions.

For locating the certificate-specific files, the generated data and axiom modules share the prefix

`CertificatePhase4ScalarN150Phase420260827HalfDC`,

with suffixes `.lean` and `Axioms.lean`, respectively. The public theorem module uses the prefix

`MainPhase4ScalarN150Phase420260827HalfDC`,

and the audit module is `AuditPhase4ScalarN150HalfDC.lean`.